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Summary

PhD candidate in Physics at Purdue University, implementing and benchmarking quantum algorithms on real hardware (IBM, trapped-ion) against experimental condensed matter data. Led the first demonstration of a quantum computer reproducing a real material's neutron-scattering spectrum, covered by the [IBM Research blog](#), [Nature News](#), etc. Experienced across error mitigation (ZNE, PEC, DD, noise learning), circuit optimization, and sample-based quantum diagonalization, with three IBM Quantum internships and cross-institution collaboration (ORNL, LANL, IBM). Proficient in Qiskit, PennyLane, CUDA-Q, and tensor network methods.

Education

Purdue University

PhD, Physics (Quantum Computing), GPA: 3.9/4.0

West Lafayette, USA

Aug 2022 – Present

Indian Institute of Science (IISc) Bengaluru

BS in Physics, CGPA: 9.3/10.0

Bengaluru, India

Aug 2018 – Jul 2022

Work Experience

IBM Quantum

Quantum Research Intern, Quantum Algorithms and Engineering

Yorktown Heights, NY

June 2026 – Present

- Contributing to research on quantum algorithm implementation and hardware benchmarking within the Quantum Algorithms and Engineering team.

IBM Quantum

Quantum Research Intern

Yorktown Heights, NY

May 2025 – July 2025

- Improved energy estimation in Sample-Based Quantum Diagonalization (SQD) subroutines for [2Fe-2S] cluster and N₂ systems, boosting accuracy and noise resilience.
- Built CPU-parallelized diagonalization routines for large quantum bitstring outputs, enabling high-throughput assessment of circuit outputs.
- Developed internal tooling adopted by the quantum chemistry research team for iterative SQD workflows on IBM HPC infrastructure.
- Co-filed a patent with IBM researchers based on algorithmic improvements developed during the internship.

IBM Quantum

Quantum Research Intern

Yorktown Heights, NY

May 2024 – Feb 2025

- Simulated superdiffusion breakdown in 2D Heisenberg spin chains on high-depth IBM hardware circuits, contributing to a Phys. Rev. B publication (2026).
- Applied Probabilistic Error Cancellation and error amplification to recover signal from deep circuits, improving quantum simulation results.
- Built an internal software package to compute two-body correlation functions, with automated testing (Travis, tox).
- Implemented transmon qutrit calibration (Rabi/Ramsey) and simulated AKLT chains to demonstrate advantages of using qutrits over qubits (arXiv:2412.19786).

Relevant Research Experience

Quantum Simulation

Graduate Research Assistant

Purdue University

Aug 2022 - Present

- Led the first demonstration of a quantum computer simulating a real material's neutron-scattering spectrum (KCuF₃)**, directly benchmarking hardware output against experimental inelastic neutron scattering data (arXiv:2603.15608); work covered by the [IBM Research blog](#), [Nature News](#), etc.
- Investigated robust chiral edge dynamics of Kitaev honeycomb lattice models on a trapped-ion processor, co-authoring arXiv:2507.08939.
- Simulated spin-1 AKLT models using calibrated qutrit gates on IBM transmons and tensor networks, demonstrating a qutrit advantage over qubit encodings (arXiv:2412.19786).

- Simulated superdiffusive breakdown in Heisenberg chains with 2D interactions on quantum hardware, using error mitigation protocols to recover signal from noisy circuits (Phys. Rev. B, 2026).
- Designed auxiliary Hamiltonians and a modular, gate-efficient ansatz for resonating valence bond ground-state simulation, achieving < 1% ground-state energy accuracy on IBMQ hardware (J. Phys. Chem. A, 2023).

Quantum Algorithms

Graduate Research Assistant

Purdue University

Aug 2022 - Present

- Developed improved sample-based quantum diagonalization (SQD) routines and variational quantum algorithms, integrating Qiskit, PennyLane, and custom tensor network methods.
- Benchmarked quantum random projection for dimensionality reduction against classical PCA on MNIST/CIFAR-100, extracting dominant singular vectors via variational quantum SVD (Phys. Rev. Research, 2024).
- Utilized Purdue's NVIDIA A30 GPUs and SLURM cluster to run large-scale quantum simulations (including spin transport and ground state circuit optimization).
- Worked with CUDA-Q libraries and open-source quantum error correction (QEC) toolkits for syndrome decoding and fault-tolerant circuit simulation workflows.

Open Source Contributions

Qutrit Calibration Toolkit & AKLT Simulations

- My [spin-1 AKLT](#) research directly led to the development and deployment of transmon qutrit calibration codes in the Qiskit Community project ([qutrit-calibration](#)).

Skills

- **Programming:** Python, R, C/C++, Matlab, Mathematica.
- **Libraries:** Pandas, PyTorch, NumPy, Scikit-learn, Keras, TensorFlow.
- **Quantum Programming:** Qiskit, PennyLane, Cirq, CUDA-Q, STIM.
- **HPC Tools:** SLURM, CUDA, tensor networks, DMRG.
- **Hardware simulation:** qutip, squbits.

Publications and Preprints

1. Benchmarking quantum simulation with neutron-scattering experiments, [arXiv:2603.15608](#), 2026
2. Robust Chiral Edge Dynamics of Kitaev Honeycomb on a Trapped Ion Processor, [arXiv:2507.08939](#), 2025
3. Superdiffusion resilience in Heisenberg Chains with 2D interactions on a quantum processor, Phys. Rev. B, 2026
4. Transmon qutrit-based simulation of spin-1 AKLT systems, [arXiv:2412.19786](#), 2025 (under review at npj Quantum Information)
5. Random projection using random quantum circuits, Phys. Rev. Res. 6, 2024
6. Physics-Inspired Quantum Simulation of RVB States, J. Phys. Chem. A, 127(41), 2023

Achievements

- Purdue Graduate Student Travel Award Recipient (2026)
- Reviewer, Journal of Physics A: Mathematical and Theoretical (2023)
- DAAD WISE Scholar (2021), KVPY Scholar (2018–22)

Mentoring, Teaching and Extracurricular Involvement

Actively mentored and taught undergraduates as **QSTEP 2025 Workshop Instructor**, served as **Teaching Assistant at Purdue University** (Aug 2021–May 2023), and participated in quantum computing competitions and training events including **MITiQuHack**, **QHack 2023** (ranked 112/800), **PennyLane Coding Camp** (ranked 59/450), **IBM Fall Quantum Challenge**, and **Quantum ML Summer School**.